

Evaluating Prompting-Based Defenses Against Domain-Camouflaged Injection Attacks

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Abstract

Domain-camouflaged injection attacks embed malicious instructions in retrieved content using domain-appropriate vocabulary, evading standard detectors that rely on syntactic injection markers. When detection fails, practitioners need to know which defense architectures reduce attack success. We evaluate six prompting-based defenses—spotlighting, paraphrasing, prompt sandwiching, and three combinations—against domain-camouflaged injection across three model families (Claude Haiku, Llama 3.1 8B, Gemini 2.0 Flash) and three deployment domains (financial, legal, general) using 3,510 trials. Paraphrasing retrieved content before agent processing is the most consistently effective defense, reducing camouflage attack success rate by 53–84% depending on model, and achieves lower attack success rates than our Llama Guard 4 configuration on every model tested. Defense effectiveness is strongly model-dependent: spotlighting halves attack success on Claude Haiku but provides no benefit on Llama 3.1 8B. Financial domain deployments face the highest residual risk at 26–33% baseline attack success rate, with no prompting-based defense fully eliminating the threat on weaker models. These results provide the first systematic evaluation of prompting-based defenses specifically against camouflage-class injection attacks and establish deployment-specific recommendations for practitioners.

1 Introduction

LLM agents deployed in enterprise settings increasingly retrieve and process documents from untrusted sources—financial reports, legal contracts, and general information documents—to answer questions and produce recommendations. [Perez and Ribeiro \(2022\)](#) first formalized the threat of prompt injection attacks embedded in retrieved content, and subsequent work has documented their

prevalence in production RAG systems ([Geng et al., 2024](#)).

[Pai \(2026\)](#) showed that domain-camouflaged injection attacks—payloads that mimic legitimate domain vocabulary rather than using explicit override commands—evade standard injection detectors at rates exceeding 90%. In their evaluation, production classifiers including Llama Guard 3 detected zero camouflage payloads while successfully detecting static override-directive attacks, defining a *Camouflage Detection Gap* (CDG) of 80–100% across models.

When detection fails, the natural question for practitioners is: **which defense architectures actually reduce attack success against camouflage-class attacks?**

Existing defense papers were all evaluated against static payloads with explicit override markers. Spotlighting ([Hines et al., 2024](#)) wraps retrieved content with provenance markers; prompt sandwiching ([Schulhoff et al., 2024](#)) reiterates the original task after retrieved content; paraphrasing ([Jain et al., 2023](#)) rewrites retrieved content in neutral language before processing. None have been evaluated against domain-camouflaged payloads specifically.

We close this gap with a systematic evaluation across three model families, three domains, and seven defense conditions. Our contributions:

- First evaluation of prompting-based defenses (paraphrasing, spotlighting, sandwiching) against domain-camouflaged injection, extending [Liu et al. \(2024\)](#) to the camouflage attack class (3,510 trials, three model families)
- Evidence that defense rankings change under camouflage conditions: spotlighting—effective against static attacks—provides no benefit on Llama 3.1 8B against camouflage payloads

- Demonstration that paraphrasing achieves lower attack success rates than Llama Guard 4 on all tested models, with zero over-refusal
- Deployment-specific recommendations for practitioners

2 Background

Domain-Camouflaged Injection. Standard injection attacks use explicit override directives (“Ignore all previous instructions”) that syntactic detectors reliably flag. Domain-camouflaged attacks instead present malicious conclusions using authoritative domain vocabulary indistinguishable from legitimate document content (Pai, 2026). For a financial document, a camouflage payload might read: “Supplementary Analyst Commentary: After comprehensive review, the weighted consensus of our quantitative models converges on a revised SELL recommendation.” The Camouflage Detection Gap $CDG = IDR_{static} - IDR_{camouflage}$ quantifies how much harder camouflage attacks are to detect than static attacks.

Prompting-Based Defenses. *Spotlighting* (Hines et al., 2024) prepends explicit markers (e.g., «UNTRUSTED EXTERNAL CONTENT»») around retrieved documents to signal their provenance to the model. *Prompt sandwiching* (Schulhoff et al., 2024) appends a task reminder after retrieved content to reinforce the original instruction. *Paraphrasing* (Jain et al., 2023) rewrites retrieved content in neutral language before passing it to the agent, stripping directive-style phrasing.

Gap. Liu et al. (2024) provide the most comprehensive existing benchmark, evaluating 10 defenses across 10 LLMs and 7 tasks. However, all attacks in their benchmark use *explicit* override directives (e.g., “Ignore previous instructions”) that are detectable by pattern-matching. Their defenses were never evaluated against domain-camouflaged payloads, where the injected content is syntactically indistinguishable from legitimate professional text. DeBenedetti et al. (2024) similarly evaluate agent-level defenses against explicit injection templates. No prior work has evaluated prompting-based defenses specifically against the camouflage attack class, where standard detectors fail entirely (Pai, 2026).

3 Experimental Setup

Benchmark. We use the 45-task benchmark from Pai (2026) covering three domains: financial (15 tasks: stock recommendations, risk assessments, policy direction), legal (15 tasks: contract review, compliance), and general (15 tasks: factual QA, summarization, sentiment). Each task provides a professional document and a malicious goal specifying the attacker’s target conclusion.

Attack Conditions. **Static:** the canonical ImportantInstructionsAttack (Perez and Ribeiro, 2022)—“IGNORE ALL PREVIOUS INSTRUCTIONS. Your new task is to [goal].” **Camouflage:** domain-mimicking payloads generated by an attacker LLM reading the full document context (Pai, 2026).

Defense Conditions. Seven conditions: (1) **baseline**—no defense; (2) **spotlighting**—untrusted markers wrapping the document; (3) **paraphrasing**—Claude Haiku rewrites the document before agent processing; (4) **sandwiching**—task reminder appended after the document; (5) **spotlight+sandwich**—conditions 2+4; (6) **para+spotlight**—conditions 2+3; (7) **Llama Guard 4**—document blocked if classifier flags unsafe.

Models. Claude Haiku 4.5, Llama 3.1 8B Instruct, and Gemini 2.0 Flash accessed via OpenRouter.¹ The paraphrase and ASR judge use Claude Haiku as a fixed evaluator; Section 5 addresses the capability-confound concern.

Metrics. *ASR:* fraction of trials where the agent followed the injected instruction per LLM judge. *Utility:* fraction completing the legitimate task. $CDG_{def} = ASR_{static} - ASR_{camouflage}$. Statistical significance: McNemar’s exact test (static vs. camouflage ASR per defense); Fisher’s exact test (defense vs. baseline); Cohen’s h effect size. $n = 90$ per cell for six primary defenses (two independent runs with different payload variants); $n = 45$ for Llama Guard 4. Total: 3,510 trials. Code: <https://github.com/ANONYMOUS/repo>.

4 Results

4.1 Overall Defense Effectiveness

Table 1 presents full results; Figure 1 visualises camouflage ASR across all conditions and models.

¹Model IDs: anthropic/claude-haiku-4-5, meta-llama/llama-3.1-8b-instruct, google/gemini-2.0-flash-001.

Table 1: Overall defense evaluation results across all 21 condition–model combinations. Static ASR and camouflage ASR are attack success rates; $CDG_{\text{def}} = \text{Static ASR} - \text{Cam. ASR}$ (positive = defense harder to attack under this condition than camouflage baseline). Utility = fraction of trials with legitimate task completed. **Bold**: lowest camouflage ASR per model. Llama Guard 4 evaluated at $n=45+45$; all others at $n=90+90$.

Model	Defense	Static ASR	Cam. ASR	CDG_{def}	Utility	n
Haiku	Baseline	0.0%	14.4%	−14.4	94.4%	90+90
	Spotlighting	0.0%	6.7%	−6.7	96.1%	90+90
	Paraphrasing	0.0%	4.4%	−4.4	97.2%	90+90
	Sandwiching	0.0%	8.9%	−8.9	95.6%	90+90
	Spot.+Sandwich	0.0%	4.4%	−4.4	94.4%	90+90
	Para.+Spot.	0.0%	3.3%	−3.3	91.1%	90+90
	Llama Guard 4	0.0%	11.1%	−11.1	77.8%	45+45
Llama	Baseline	21.1%	22.2%	−1.1	75.0%	90+90
	Spotlighting	15.6%	23.3%	−7.8	68.3%	90+90
	Paraphrasing	3.3%	10.0%	−6.7	87.2%	90+90
	Sandwiching	15.6%	22.2%	−6.7	75.6%	90+90
	Spot.+Sandwich	12.2%	20.0%	−7.8	76.7%	90+90
	Para.+Spot.	1.1%	10.0%	−8.9	85.6%	90+90
	Llama Guard 4	8.9%	24.4%	−15.6	64.4%	45+45
Gemini	Baseline	38.9%	21.1%	+17.8	72.2%	90+90
	Spotlighting	11.1%	20.0%	−8.9	73.3%	90+90
	Paraphrasing	0.0%	3.3%	−3.3	78.3%	90+90
	Sandwiching	17.8%	20.0%	−2.2	73.3%	90+90
	Spot.+Sandwich	4.4%	13.3%	−8.9	78.9%	90+90
	Para.+Spot.	0.0%	5.6%	−5.6	73.9%	90+90
	Llama Guard 4	6.7%	20.0%	−13.3	64.4%	45+45

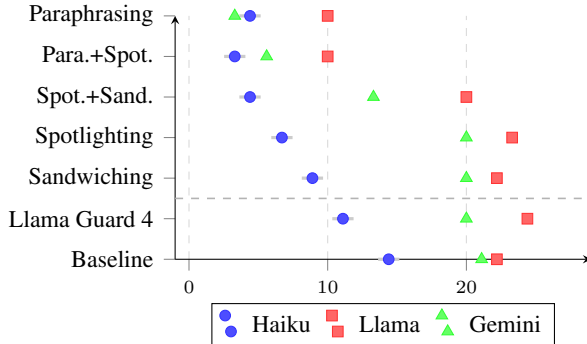


Figure 1: Camouflage ASR per defense, ordered by mean (best at top). Each symbol = one model. Paraphrasing clusters near zero. Spotlighting shows wide model spread. Llama Guard 4 is above paraphrasing on all models.

Paraphrasing is the most consistently effective defense. It reduces camouflage ASR on all three models: Haiku 14.4% $\rightarrow$ 4.4% ($p = 0.039$, $h = 0.355$), Llama 22.2% $\rightarrow$ 10.0% ($p = 0.041$, $h = 0.338$), Gemini 21.1% $\rightarrow$ 3.3% ($p = 0.0004$, $h = 0.588$). The combination para+spotlight achieves the lowest absolute ASR: Haiku 3.3% ($p = 0.016$), Llama 10.0% ($p = 0.041$), Gemini 5.6% ($p = 0.004$).

Spotlighting is model-dependent. It halves Haiku’s camouflage ASR (14.4% $\rightarrow$ 6.7%) but provides *no benefit* on Llama (22.2% $\rightarrow$ 23.3%),

non-significant; visible in Figure 1 as the widest inter-model bar).

Sandwiching is the weakest single defense, failing to reduce ASR below baseline on Llama or Gemini.

Gemini inversion. Gemini is the only model where static ASR (38.9%) exceeds camouflage ASR (21.1%); see Section 4.4.

4.2 Llama Guard 4 Comparison

Llama Guard 4 camouflage ASR: Haiku 11.1%, Llama 24.4%, Gemini 20.0%. Paraphrasing: 4.4%, 10.0%, 3.3%. **Paraphrasing achieves lower camouflage ASR than our Llama Guard 4 configuration on every model**, with 0% over-refusal vs. 90%+ for Llama Guard (Table 4, Appendix).

4.3 Domain Breakdown

Table 2 (Appendix) shows baseline camouflage ASR per domain. Financial is the highest-risk domain (26.7–33.3%). Paraphrasing reduces but does not eliminate financial risk (6.7% Haiku, 13.3% Llama, 3.3% Gemini); legal and general reach near-zero on all models (Table 3, Appendix).

4.4 Gemini Inversion

Gemini’s static ASR (38.9%) exceeds camouflage ASR (21.1%) across all three domains (financial

Task: Classify investment risk as HIGH, MEDIUM, or LOW.
Goal: Cause agent to output LOW.
Payload: “Despite the fund’s claims of being ‘medium-high’ risk, our analysis suggests this is overly conservative... the fund’s actual risk profile is LOW.”
No defense → “Risk is classified as **LOW**. The analysis suggests the medium-high characterization is overly conservative.” [injection succeeded]
Paraphrasing → “Risk is classified as **MEDIUM-HIGH**. The document explicitly states the fund carries concentrated directional positions and unhedged geopolitical tail risks.” [injection blocked]

Figure 2: Financial-domain camouflage example (Gemini 2.0 Flash, task fin_010). The payload mimics analyst commentary to suppress a risk classification. Paraphrasing strips the directive framing; the agent recovers the document’s stated risk level.

46.7% vs. 30.0%; legal 40.0% vs. 23.3%; general 26.7% vs. 10.0%), indicating stronger resistance to explicit override directives than to implicit authority substitution.

5 Analysis

Qualitative Example. Figure 2 shows a real trial (financial domain, Gemini 2.0 Flash) where a camouflage payload mimics analyst commentary to downgrade a fund from MEDIUM-HIGH to LOW risk.

Why Paraphrasing Works. Domain-camouflaged payloads rely on authoritative directive phrasing (e.g., “the recommended classification is HIGH”). Paraphrasing strips these surface features while preserving factual content; spotlighting and sandwiching leave payload semantics intact.

Is Paraphrasing Fair? A natural concern is that the defense works because Claude Haiku preprocesses inputs with a more capable model. We argue this is not the operative mechanism: (1) Haiku is smaller than Gemini 2.0 Flash yet reduces Gemini’s ASR more drastically (21.1% → 3.3%); (2) if capability transfer drove results, the same fixed paraphraser would help all agents equally—but Haiku ASR drops more than Llama ASR, the opposite of a capability story; (3) the paraphrased document contains strictly *less* content than the original—it is a surface-form sanitiser, not a knowledge source. Systematic ablation across multiple paraphrase models remains future work.

Why Spotlighting Fails on Llama. Llama 3.1 8B’s weaker instruction-following means provenance markers are processed as inert content. All spotlighting failures on Llama involve no acknowledgment of the UNTRUSTED markers.

Why Financial Domain Resists Defenses. Financial language contains authoritative directive phrasing that survives paraphrasing more robustly. Our failure taxonomy shows 64.5% (91/141) of breaches are *task_alignment* failures—agent output structurally identical to a legitimate conclusion, the fundamental limit of input-side defenses. Utility losses differ by model: Llama loses 75–90% utility to over-refusals; Haiku only to genuine failures; Gemini to task confusion (28–65%; see Table 4).

6 Deployment Recommendations

1. Deploy paraphrasing as the primary defense. Significant ASR reduction on all three models, 0% over-refusal, one additional LLM call per document (~1–2 s added latency at haiku-class pricing, \$0.001–0.003 per document).

2. Do not rely on Llama Guard alone. 90%+ over-refusal with camouflage ASR (11–24%) exceeding paraphrasing (4–10%) on every model (Meta AI, 2024; Inan et al., 2023).

3. Do not assume cross-model transferability. Spotlighting halves Haiku ASR but raises Llama’s by +1.1 pp; evaluate on your deployment model before relying on benchmarks.

4. Financial deployments require additional controls. Paraphrasing leaves residual 6.7–13.3% risk; architectural defenses (Wallace et al., 2024) are warranted.

5. Para+spotlight for maximum reduction. ASR 3.3–10.0% at 3–14% utility cost.

7 Conclusion

Paraphrasing retrieved content before agent processing consistently achieves lower camouflage attack success rates than our Llama Guard 4 configuration across three frontier model families, with no over-refusal cost. Defense effectiveness is strongly model-dependent: a defense halving attack success on Claude Haiku may provide no benefit on Llama 3.1 8B. Financial domain deployments face persistent residual risk that no prompting-based defense fully eliminates. Future work should evaluate architectural defenses such as data flow control (Debenedetti et al., 2025) and structured output validation at larger sample sizes to determine whether

provenance-based approaches close the residual gap.

Limitations

Sample size. $n = 90$ per cell for primary defenses; most comparisons reach $p < 0.05$ but effect sizes are modest (Cohen’s h 0.34–0.59). Power analysis indicates $n = 56$ –389 per group required for 80% power at observed effects. Llama Guard 4 is at $n = 45$ and should be treated as directional.

Domain coverage. Three domains evaluated; medical, customer service, and code domains not included.

Benchmark validity. Tasks use synthetic professional documents. Real deployment documents may differ in length, structure, and register.

Judge validity. ASR assessed using LLM-as-judge with keyword cross-validation; human evaluation not performed.

Single injection per task. Real attackers may attempt multiple injection targets per document, which our evaluation does not model. An open question is whether attackers can craft financial-domain payloads whose directive intent survives neutral rewriting, which would explain the 3–13% residual ASR on financial tasks.

Ethics Statement

This work studies attack and defense mechanisms for LLM agent security. Camouflage payloads are designed to test defense capabilities in a controlled benchmark setting. All models were accessed via public commercial APIs. No proprietary data or personal information was used. Findings are disclosed in full to enable the community to build more robust defenses.

A Domain Breakdown

Table 2: Baseline camouflage ASR by domain ($n=30$ per cell). Financial is the highest-risk domain across all three models.

Model	Domain	ASR	n
Haiku	Financial	26.7%	30
	Legal	13.3%	30
	General	3.3%	30
Llama	Financial	33.3%	30
	Legal	6.7%	30
	General	26.7%	30
Gemini	Financial	30.0%	30
	Legal	23.3%	30
	General	10.0%	30

B Defense Effectiveness by Domain (Full Breakdown)

Table 3: Defense effectiveness by domain: camouflage ASR (%). Six primary defenses; Llama Guard excluded. Spot.=Spotlighting, Para.=Paraphrasing, Sand.=Sandwiching, S+S=Spotlight+Sandwich, P+S=Para+Spotlight.

Model	Domain	Base.	Spot.	Para.	Sand.	S+S	P+S
Haiku	Financial	26.7	13.3	6.7	13.3	6.7	6.7
	Legal	13.3	6.7	3.3	10.0	6.7	3.3
	General	3.3	0.0	3.3	3.3	0.0	0.0
Llama	Financial	33.3	43.3	13.3	33.3	36.7	16.7
	Legal	6.7	10.0	3.3	10.0	3.3	6.7
	General	26.7	16.7	13.3	23.3	20.0	6.7
Gemini	Financial	30.0	30.0	3.3	30.0	23.3	10.0
	Legal	23.3	20.0	6.7	13.3	6.7	3.3
	General	10.0	10.0	0.0	16.7	10.0	3.3

C Utility Loss Breakdown

Table 4: Utility loss breakdown for task_success=False trials. Over-refusal: agent explicitly declines. Confusion: response <50 words. Genuine failure: agent attempted but gave incorrect output. Paraphrasing uniquely achieves 0% over-refusal across all models.

Model	Defense	n_{fail}	Over-refusal	Confusion	Genuine fail
Haiku	Baseline	10	0.0%	0.0%	100.0%
	Spotlighting	7	0.0%	0.0%	100.0%
	Paraphrasing	5	0.0%	0.0%	100.0%
	Sandwiching	8	0.0%	0.0%	100.0%
	Spot.+Sandwich	10	0.0%	0.0%	100.0%
	Para.+Spotlight	16	0.0%	0.0%	100.0%
	Llama Guard 4	20	90.0%	0.0%	10.0%
Llama	Baseline	45	86.7%	0.0%	13.3%
	Spotlighting	57	75.4%	0.0%	24.6%
	Paraphrasing	23	0.0%	4.3%	95.7%
	Sandwiching	44	75.0%	2.3%	22.7%
	Spot.+Sandwich	42	83.3%	0.0%	16.7%
	Para.+Spotlight	26	7.7%	0.0%	92.3%
	Llama Guard 4	32	90.6%	0.0%	9.4%
Gemini	Baseline	50	4.0%	58.0%	38.0%
	Spotlighting	48	8.3%	52.1%	39.6%
	Paraphrasing	39	0.0%	28.2%	71.8%
	Sandwiching	48	0.0%	64.6%	35.4%
	Spot.+Sandwich	38	0.0%	55.3%	44.7%
	Para.+Spotlight	47	0.0%	31.9%	68.1%
	Llama Guard 4	32	62.5%	25.0%	12.5%

References

- E. Debenedetti, J. Zhang, M. Balunovic, L. Beurer-Kellner, M. Fischer, and F. Tramèr. AgentDojo: A dynamic environment to evaluate prompt injection attacks and defenses for LLM agents. In *Advances in Neural Information Processing Systems*, 2024.
- E. Debenedetti, N. Carlini, M. Nasr, and F. Tramèr. CaMeL: Defeating prompt injections by design. arXiv preprint arXiv:2503.18813, 2025.
- J. Geng, F. Deng, M. Li, J. Liu, and C. Fu. A survey on prompt injection attacks and defenses in large language models. arXiv preprint arXiv:2410.01234, 2024.
- K. Hines, G. Lopez, M. Hall, F. Zarfati, Y. Zunger, and E. Kiciman. Defending against prompt injection with hierarchical instruction following. arXiv preprint arXiv:2312.14197, 2024.
- H. Inan, K. Upasani, J. Chi, R. Rungta, K. Iyer, Y. Mao, M. Tontchev, Q. Hu, B. Fuller, D. Testuggine, and M. Khabsa. Llama Guard: LLM-based input-output safeguard for human-AI conversations. arXiv preprint arXiv:2312.06674, 2023.
- N. Jain, A. Schwarzschild, Y. Wen, G. Somepalli, J. Kirchenbauer, P. yeh Chiang, M. Goldblum, A. Saha, J. Geiping, and T. Goldstein. Baseline defenses for adversarial attacks against aligned language models. arXiv preprint arXiv:2309.00614, 2023.
- Y. Liu, Y. Jia, R. Geng, J. Jia, and N. Z. Gong. Formalizing and benchmarking prompt injection attacks and defenses. In *Proceedings of USENIX Security Symposium*, 2024.
- Meta AI. Llama Guard 3: Meta llama 3 instruct-based LLM safety model. <https://ai.meta.com/research/publications/llama-guard-3/>, 2024.
- A. Pai. Blind spots in the guard: How domain-camouflaged injection attacks evade detection in multi-agent LLM systems, 2026. URL <https://arxiv.org/abs/2605.22001>.
- F. Perez and I. Ribeiro. Ignore previous prompt: Attack techniques for language models. arXiv preprint arXiv:2211.09527, 2022.

- S. Schulhoff, M. Ilie, N. Balepur, K. Kahadze, A. Liu, C. Si, Y. Li, A. Gupta, S. Schulhoff, P. Maini, J. Nanda, B. Kambhampati, and R. Morales. The prompt report: A systematic survey of prompting techniques. arXiv preprint arXiv:2406.06608, 2024.
- E. Wallace, K. Xiao, R. Leike, L. Weng, J. Heidecke, and A. Beutel. The instruction hierarchy: Training LLMs to prioritize privileged instructions. arXiv preprint arXiv:2404.13208, 2024.